Proton Pump Inhibitors in Infants: Why the Trials Failed, What the Drugs Cost, and When They Are Right.

Picture a seven-week-old boy brought to clinic by exhausted parents. He cries for about three hours every evening, arches his back after feeds, and brings up small amounts of milk several times a day. He is gaining weight along the fiftieth centile. A relative has told the family that he has silent reflux, and they ask whether he could have the medicine that worked for their friend's baby. It would take less than a minute to write a prescription for omeprazole. The question of this chapter is whether that prescription would help him, whether it could harm him, and whether there is ever a good reason to start a proton pump inhibitor in an infant.

The main idea is straightforward to state and surprisingly hard to practise. Regurgitation in infancy is a problem of volume and immaturity, not of acid. Proton pump inhibitors, usually shortened to PPIs, reliably raise the pH inside the stomach, and in every randomised trial in infants they have failed to reduce crying or spitting any more than a placebo. Meanwhile, the change they do produce, a less acidic stomach, is associated with more infections, more fractures and more allergy, and in preterm infants with necrotising enterocolitis and death. There are real indications, mainly erosive oesophagitis and repaired oesophageal atresia, and in those infants the drug should be given properly and stopped on purpose.

A note on sources. The main guideline is the 2018 joint clinical practice guideline on paediatric gastro-oesophageal reflux written by Rosen and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. The second is guideline NG1 from the National Institute for Health and Care Excellence in England, known as NICE, first published in 2015 and updated in 2019. For preterm infants, the American Academy of Pediatrics published a clinical report by Eichenwald in 2018. The key trials are those of Moore and colleagues in 2003 and Orenstein and colleagues in 2009, together with a review by reviewers at the United States Food and Drug Administration, published by Chen and colleagues in 2012, and a 2015 systematic review by Gieruszczak-Bialek, Szajewska and colleagues. The safety data come from a French national cohort published by Lassalle and colleagues in 2023, a United States military health system cohort on fractures published by Malchodi and colleagues in 2019, and another on allergy published by Mitre and colleagues in 2018, and from two neonatal studies, by Terrin and colleagues in 2012 and Guillet and colleagues in 2006. The American College of Gastroenterology and the American Gastroenterological Association do not have guidance on infants.

Part one. Why the drug misses the target.

Begin with the physiology, step by step. The first step is the transient relaxation of the lower oesophageal sphincter. This is a normal reflex in which the muscle ring at the bottom of the oesophagus relaxes for a few seconds, independently of swallowing, so that the stomach can vent gas. In an infant it also lets milk travel upward. The second step is volume. An infant drinks, relative to body size, a very large volume of milk, has a short oesophagus, and spends much of the day lying down. Put those together and some of the reflux becomes visible as regurgitation. The third step is buffering. Milk neutralises stomach acid, so for much of the time after a feed the stomach contents are only weakly acidic. In impedance studies, which detect the movement of fluid in the oesophagus regardless of its acidity, most reflux in young infants after feeds is weakly acidic rather than strongly acidic.

The fourth step is where a proton pump inhibitor acts. It blocks the final step of acid secretion in the parietal cells of the stomach, so the refluxed fluid becomes less acidic. But it does not reduce the number of reflux events, how high they travel, or how much volume comes up. A baby on a proton pump inhibitor spits just as often; the milk simply has a higher pH. The fifth step is the cry. Infant crying follows its own developmental curve, peaking at about six weeks of age and falling after the second month, and in most infants it does not track acid exposure at all.

Several pharmacological details make the problem worse. Proton pump inhibitors block only pumps that are active, so they work best when taken about thirty minutes before a meal, which is hard to arrange in an infant who feeds on demand every two to three hours. Their full effect takes several days of regular dosing, so an apparent response within forty-eight hours is almost certainly natural history or a placebo effect. They are cleared by a liver enzyme called CYP2C19, whose activity matures over infancy and varies with genotype, so the drug exposure at a fixed dose per kilogram is unpredictable. And what remains, after all of this, is a higher gastric pH, which is exactly the change that drives the harms we will meet later.

Part two. A drug in search of a disease.

Despite this, prescribing rose steeply. The Food and Drug Administration reviewers reported that proton pump inhibitor use in infants under twelve months in the United States rose about elevenfold between 2002 and 2009, and that in 2009 about one hundred and forty-five thousand infants received about four hundred and four thousand prescriptions.

The tide has turned, at least in some places. A 2025 study by Hirsch and colleagues looked at nearly fifty thousand children under two years of age with a diagnosis of reflux or a feeding disorder at a single United States centre between 2014 and 2023. The deck shows this as a bar chart. Proton pump inhibitor prescribing fell from about twenty-five percent of these children in 2014 to about seven percent in 2023. Prescribing of histamine type two receptor antagonists, usually called H2 receptor antagonists, the older class that includes ranitidine and famotidine, fell from seventeen percent to about eleven percent. Overall, any reflux drug was prescribed to about thirty-seven percent in 2014 and about nineteen percent in 2023. The

guideline is working, but about one in fourteen infants with these diagnoses still received a proton pump inhibitor.

Part three. Fifty-four percent versus fifty-four percent.

The landmark trial is the study by Orenstein, Hassall and colleagues published in 2009. It enrolled one hundred and sixty-two infants aged one to twelve months whose symptoms, which had been attributed to reflux disease, persisted despite at least a week of conservative management. They were randomised to lansoprazole or placebo for four weeks. The primary outcome was a reduction of at least fifty percent in feeding-related crying. The deck shows the result as a bar chart, and the first two bars are the same height. Forty-four of eighty-one infants responded in each group, which is fifty-four percent on lansoprazole and fifty-four percent on placebo. There was no difference in any secondary outcome, including regurgitation, arching, refusal of feeds and global assessments.

The harms were not equal. Sixty-two percent of infants on lansoprazole had at least one adverse event, compared with forty-six percent on placebo, a difference that just missed statistical significance. Serious adverse events, mostly lower respiratory tract infections, occurred in ten infants on lansoprazole and two on placebo, about twelve percent versus two and a half percent, and this difference was significant. That corresponds to a number needed to harm of about ten: for every ten infants treated for four weeks, one extra infant had a serious adverse event.

An earlier trial asked an even sharper question. Moore and colleagues in 2003 studied thirty irritable infants aged three to twelve months, every one of whom had proven reflux disease, either an abnormal oesophageal pH study or oesophagitis on biopsy. These were the infants most likely to benefit. Each received two weeks of omeprazole and two weeks of placebo in random order. On omeprazole, the oesophageal acid index, the proportion of time with acid in the oesophagus, fell significantly. But crying and fussing time and irritability scores were the same on drug and placebo, and crying fell in both periods, simply because the infants were getting older. The probe improved; the baby did not. If acid suppression fails in infants with proven acid reflux, it cannot be expected to succeed in the unselected crying infant.

The rest of the evidence agrees. Three randomised withdrawal trials, with pantoprazole by Winter and colleagues, rabeprazole by Hussain and colleagues, and esomeprazole by Winter and colleagues, gave the drug openly and then randomised the apparent responders to continue the drug or switch to placebo. In none was there a difference in symptoms or weight after the switch. When the Food and Drug Administration reviewed the four randomised infant trials submitted to support labelling, none showed efficacy for symptoms of reflux disease, and the reviewers advised limiting proton pump inhibitors in infants to endoscopically proven oesophagitis. The 2015 systematic review concluded that the data do not support using proton pump inhibitors to reduce infant crying and irritability. It is worth remembering that most of these trials were funded by industry and designed to find an effect.

Part four. What the drugs cost.

Gastric acid is part of the immune system. It kills swallowed bacteria and viruses, and it shapes the microbes that colonise the gut. The largest study of what happens when that barrier is lowered is the French national cohort published in 2023, which followed one million two hundred and sixty thousand children born between 2010 and 2018, of whom about six hundred and seven thousand were exposed to a proton pump inhibitor, at a median age of eighty-eight days. The deck shows the results as a forest plot of adjusted hazard ratios. Overall, serious infections were about thirty-four percent more common after exposure, with a hazard ratio of one point three four and a very narrow confidence interval. Bacterial infections had the highest ratio, at one point five six, followed by infections of the digestive tract at one point five two and of the ear, nose and throat at one point four seven. Infections of the nervous system had a ratio of one point three one, viral infections one point three, lower respiratory infections one point two two, and kidney and urinary infections one point two. This is an observational study, so some confounding by indication is likely, because sicker children are more often given the drug. But the direction matches the excess of respiratory infections in the randomised trial, and an earlier Italian study by Canani and colleagues found pneumonia in twelve percent of children treated with acid suppression for eight weeks, compared with two percent of healthy controls, over four months.

The second cost is to bone. In the Malchodi study of eight hundred and fifty-one thousand children in the United States Military Health System, about eleven percent received acid suppression in the first year of life. After adjustment, proton pump inhibitor use alone was associated with about a twenty-one percent increase in the hazard of fracture in childhood, and combined use with an H2 receptor antagonist with about a thirty percent increase. H2 receptor antagonists alone were not associated with fracture. The median age at first fracture was three point nine years in exposed children and four point five years in unexposed children, and the hazard rose with more days of use and with earlier first use. A plausible mechanism is reduced calcium absorption at a higher gastric pH during the years of fastest bone growth.

The third cost is allergy. Pepsin, the enzyme that begins protein digestion in the stomach, needs acid to work. At a higher pH, dietary proteins reach the intestine less digested and therefore more able to sensitise the immune system. In the Mitre study of seven hundred and ninety-two thousand children, acid suppression in the first six months was associated with later allergic disease. The deck shows the hazard ratios as horizontal bars. For food allergy the ratio was about two point six after a proton pump inhibitor and about two point two after an H2 receptor antagonist. For medication allergy it was about one point eight and one point seven. For anaphylaxis it was about one point five for each class, and for allergic rhinitis about one point four to one point five. Proton pump inhibitor use beyond sixty days carried more food allergy than shorter use. There is an uncomfortable irony here. Cow's milk protein allergy can look exactly like reflux, and a drug given for presumed reflux may help to create

the very allergy that mimics it.

The fourth cost falls on the most vulnerable infants. In a multicentre prospective study of two hundred and seventy-four very low birth weight infants published by Terrin and colleagues in 2012, those given ranitidine had infections in about thirty-seven percent, compared with about ten percent of unexposed infants, an odds ratio of about five and a half. Necrotising enterocolitis was about six and a half times more likely, and mortality was about ten percent compared with under two percent. Guillet and colleagues, in more than eleven thousand very low birth weight infants in the National Institute of Child Health and Human Development network, found that infants who received H2 blockers were about one point seven times more likely to develop necrotising enterocolitis. The American Academy of Pediatrics clinical report on reflux in preterm infants states that reflux in this group is normal and resolves with age, that acid-blocking drugs lack evidence of efficacy, and that significant harm has been associated with their use.

Part five. When a proton pump inhibitor is the right drug.

None of this means the drug has no place. The rule is to treat a lesion, not a behaviour. The clearest indication is erosive oesophagitis seen at endoscopy. NICE says to offer a proton pump inhibitor or an H2 receptor antagonist in that situation, and esomeprazole is labelled in the United States for erosive oesophagitis in infants from one month of age. The second is repaired oesophageal atresia. The joint ESPGHAN and NASPGHAN guideline on oesophageal atresia, published by Krishnan and colleagues in 2016, recommends starting a proton pump inhibitor in the neonatal period and continuing it to the end of the first year or longer if reflux disease persists, while acknowledging that the evidence is of low quality. The third is genuine acid injury, such as haematemesis with oesophagitis or a peptic stricture, which is not the same as a few brown flecks in a spit-up.

When the drug is indicated, give it properly. A typical dose of omeprazole is about one milligram per kilogram per day, once daily, before a feed. Set a duration from the start, for example eight to twelve weeks for erosive oesophagitis, and plan the review. And before any proton pump inhibitor, re-check the red flags, the growth chart and the feeding history, and consider cow's milk protein allergy, which is properly tested by two to four weeks of an extensively hydrolysed formula followed by a planned re-challenge.

Part six. A trial is not a test.

Many clinicians justify an empirical trial with the reasoning that if the baby gets better, it must have been reflux. Four facts undermine that logic. First, fifty-four percent of infants responded to placebo in four weeks in the Orenstein trial, so any response has to clear that bar. Second, crying peaks at around six weeks and declines after the second month whatever is given. Third, about ninety percent of infants with frequent regurgitation are better by twelve months. Fourth, parents usually seek help during the worst week, and the following

week is usually better, which is the statistical phenomenon called regression to the mean.

Here the two main guidelines differ. The joint NASPGHAN and ESPGHAN guideline suggests not using a trial of a proton pump inhibitor as a diagnostic test for reflux disease in infants, and reserves a four to eight week trial for typical symptoms such as heartburn in older children. NICE, in contrast, allows a four-week trial of a proton pump inhibitor or an H2 receptor antagonist in infants and children who cannot describe their symptoms, when there are unexplained feeding difficulties, distressed behaviour or faltering growth, with review at four weeks and referral if symptoms do not resolve or recur after stopping. If you do start a trial, write down in advance the outcome you expect, the date you will stop, and what you will do if symptoms return, and then stop on that date.

Part seven. Stop at the review.

Deprescribing deserves as much attention as prescribing. At every visit, find every infant on a proton pump inhibitor and ask four questions: what is the indication, when was it started, is the dose still right for the weight, and who will review it. Re-label the problem for the family, explaining the natural history and calling it reflux rather than disease. If there is no indication, stop at the review. After long use, many clinicians halve the dose for one to two weeks before stopping, although this is expert practice rather than tested advice. Warn the family to expect a brief increase in symptoms. In a study of one hundred and twenty healthy adult volunteers published by Reimer and colleagues in 2009, forty-four percent developed new acid-related symptoms after eight weeks of esomeprazole was stopped, compared with fifteen percent after placebo, which shows that rebound acid secretion is real. Reassuringly, the infant withdrawal trials found no worsening when responders were switched to placebo. NICE applies the same discipline to thickened feeds, which should not be continued indefinitely, and advises trying to stop treatments at intervals to see whether the infant has outgrown the problem.

Part eight. What the guidelines say.

Comparing the joint guideline with NICE gives a mostly aligned picture with one material difference. On visible regurgitation in an otherwise healthy infant, they are aligned: neither recommends acid suppression. On erosive oesophagitis, they are aligned: treat it. On prokinetic drugs such as metoclopramide and domperidone, they are aligned: the joint guideline recommends against them, and NICE allows them only with specialist agreement when the benefit outweighs the risk. On antacids and alginates they differ: the joint guideline does not recommend them for chronic treatment, while NICE allows a one to two week alginate trial after feeding changes have failed. On the infant who cries or seems distressed, the difference is material. The joint guideline recommends not using an H2 receptor antagonist or a proton pump inhibitor for crying or distress in otherwise healthy infants, while NICE allows distressed behaviour, alongside feeding difficulty or faltering growth, to justify a four-week trial. For preterm infants, both are silent, and the American Academy of

Pediatrics fills the gap by advising against acid blockade. And on how to stop, there is a gap: neither guideline gives a deprescribing protocol.

What to do differently in clinic.

Return to the seven-week-old boy from the start of this chapter. He is thriving, so he does not need a proton pump inhibitor. He needs an explanation of the crying curve, a careful look at his feeding volume and technique, and a review in two weeks. A three-month-old who has been on omeprazole since a two-week admission, whose parents say it worked, has no documented indication; stop it now, and explain the placebo response. A four-month-old with eczema, blood-streaked stools, spitting and slow weight gain should be assessed for cow's milk protein allergy, with two to four weeks of an extensively hydrolysed formula, not a proton pump inhibitor. A former twenty-eight-week infant in the neonatal unit with desaturations after feeds should not receive acid blockade; reflux does not cause most of these events, and the harm in this group is documented. An infant on day ten after repair of oesophageal atresia can reasonably start a proton pump inhibitor and continue it to twelve months or longer if reflux disease persists, with a review at every visit. And an infant with haematemesis and ulcerative oesophagitis at endoscopy should be treated with omeprazole at about one milligram per kilogram per day for eight to twelve weeks, and then reassessed.

So tomorrow in clinic, remember fifty-four percent versus fifty-four percent. Do not use acid suppression for crying, distress or visible spitting in an otherwise healthy infant, and do not use a proton pump inhibitor trial as a diagnostic test. Remember the costs: about a third more serious infections, more fractures, more than twice the food allergy, and in the preterm infant, necrotising enterocolitis and death. Treat a lesion when you see one, in erosive oesophagitis and after oesophageal atresia repair, with a proper dose and a stop date. Look for milk protein allergy before reaching for acid suppression. And at every visit, ask why the infant is still on the drug, and stop it at the review.